Math 230B Lecture Notes

Lance Remigio

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Contents

1 Week 5

Example ($SL(n,\mathbb{F})$). Suppose $\mathbb{F}=\mathbb{R}$ (or $\mathbb{C}$). We claim that

$$\mathrm{SL}(n,\mathbb{F})=\{A\in\mathrm{GL}(n,\mathbb{F}):\det A=1\}$$

is a Lie subgroup of $\mathrm{GL}(n,\mathbb{F})$.

Indeed, we have that

- $\mathrm{SL}(n,\mathbb{F})$ is closed under multiplication, indeed, if $A, B \in \mathrm{SL}(n,\mathbb{F})$, then

$$\det(AB) = (\det A)(\det B) = 1 \cdot 1 = 1.$$

So, AB is invertible and $\det(AB) = 1$. This implies that

$$AB \in SL(n,\mathbb{F}).$$

- $SL(n,\mathbb{F})$ contains the inverse of each of its elements. Indeed, suppose $A \in SL(n,\mathbb{F})$, then A is invertible and

$$\det(A^{-1}) = \frac{1}{\det A} = \frac{1}{1} = 1.$$

So, $A^{-1} \in \mathrm{SL}(n,\mathbb{F})$.

From the above, we can conclude that $\mathrm{SL}(n,\mathbb{F})$ is a subgroup of $\mathrm{GL}(n,\mathbb{F})$.

2 Week 6

Example (Special Orthogonal Group)**.** Note that

$$\begin{aligned}\mathrm{SO}(n) &= \{A \in M(n,\mathbb{R}) : A^T A = I \text{ and } \det A > 0\} \\ &= \{A \in M(n,\mathbb{R}) : A^T A = I \text{ and } \det A = 1\}\end{aligned}$$

We will prove that $\mathrm{SO}(n)$ is a subgroup of $\mathrm{O}(n)$.

- Note that $\mathrm{SO}(n)$ is closed under multiplication. Indeed, suppose A and B are in $\mathrm{SO}(n)$. Our goal is to show that $AB \in \mathrm{SO}(n)$. Then we have

$$A, B \in \mathrm{SO}(n) \implies A, B \in \mathrm{O}(n) \implies AB \in \mathrm{O}(n).$$

Since $A, B \in \mathrm{SO}(n)$, it follows that $\det A = 1$ and $\det B = 1$. Hence,

$$\det AB = (\det A)(\det B) = 1 \cdot 1 = 1.$$

- Also observe that $\mathrm{SO}(n)$ contains the inverse of each of its elements. Indeed, if $A \in \mathrm{SO}(n)$, then

$$A \in \mathrm{SO}(n) \implies A \in \mathrm{O}(n) \implies A^{-1} \in \mathrm{O}(n). \tag{1}$$

Furthermore,

$$A \in \mathrm{SO}(n) \implies \det A = 1 \implies \det A^{-1} = \frac{1}{1} = 1 \implies A^{-1} \in \mathrm{SO}(n) \tag{2}$$

From (1) and (2), we can see that $\mathrm{SO}(n)$ is a subgroup of $\mathrm{O}(n)$. Now, we need to find out whether $\mathrm{SO}(n)$ is a Lie subgroup of $\mathrm{O}(n)$. Note that $\mathrm{SO}(n) = \mathrm{O}(n) \cap \det^{-1}(0,\infty)$ where $(0,\infty)$ is an open set, det is a continuous function, and the preimage of an open set will be open in $M(n,\mathbb{R})$. Hence, $\mathrm{SO}(n)$ will be open in $\mathrm{O}(n)$. Thus, $\mathrm{SO}(n)$ is an embedded submanifold of $\mathrm{O}(n)$ with dimension $\frac{n^2-n}{2}$. Thus, $\mathrm{SO}(n)$ is a Lie subgroup of $\mathrm{O}(n)$ with dimension $\frac{n^2-n}{2}$.

Example (Special Unitary Group)**.** Recall that

$$\mathrm{SU}(n) = \{A \in M(n,\mathbb{C}) : A^* A = I \text{ and } \det A = 1\}.$$

We will show that the above is also a subgroup of $\mathrm{U}(n)$. Clearly, we can see that $\mathrm{SU}(n) \subseteq \mathrm{U}(n)$.

- $\mathrm{SU}(n)$ is closed under multiplication. Indeed, suppose A and B are in $\mathrm{SU}(n)$. Then observe that

$$A, B \in \mathrm{SU}(n) \implies A, B \in \mathrm{U}(n) \implies AB \in \mathrm{U}(n) \tag{1}$$

and

$$\begin{aligned} A, B \in \mathrm{SU}(n) &\implies \det A = 1 \text{ and } \det B = 1 \\ &\implies \det AB = 1 \\ &\implies AB \in \mathrm{SU}(n). \end{aligned}$$

- Also, $\mathrm{SU}(n)$ contains the inverse of each of its elements. Suppose $A \in \mathrm{SU}(n)$. Then

$$A \in \mathrm{SU}(n) \implies A \in \mathrm{U}(n) \implies A^{-1} \in \mathrm{U}(n) \tag{3}$$

and

$$A \in \mathrm{SU}(n) \implies \det A = 1 \implies \det A^{-1} = \frac{1}{\det A} = \frac{1}{1} = 1 \tag{4}$$

Now, (3) and (4) imply that $A^{-1} \in \mathrm{SU}(n)$.

Now, we need to find out if $\mathrm{SU}(n)$ a Lie subgroup of $\mathrm{U}(n)$? Of course! Let $F : \mathrm{U}(n) \to \mathbb{R}$ be defined as follows:

$$F(A) = \varphi.$$

where we previously proved that if $A \in \mathrm{U}(n)$, then $|\det A| = 1$. So, there exists an angle $\varphi \in (-\pi, \pi]$ such that $\det A = e^{i\varphi}$. It can be shown that

$$\forall A \in \mathrm{U}(n) \quad \operatorname{rank} DF(A) = 1.$$

Thus, $\mathrm{SU}(n) = F^{-1}(0)$ is an $n^2 - 1$ dimensional embedded submanifold of $\mathrm{U}(n)$. Therefore, $\mathrm{SU}(n)$ is a Lie subgroup of $\mathrm{U}(n)$ with dimension $n^2 - 1$.

Definition (Bilinear Form). A function $B : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ is said to be a **bilinear form** on $\mathbb{R}^n$ if it satisfies the following condition:

(1) $\forall x, y, z \in \mathbb{R}^n \; B(x + y, z) = B(x, z) + B(y, z)$

(2) $\forall c \in \mathbb{R}$ and $\forall x, y \in \mathbb{R}^n \; B(cx, y) = cB(x, y)$

(3) $\forall x, y, z \in \mathbb{R}^n \; B(x, y + z) = B(x, y) + B(x, z)$

(4) $\forall c \in \mathbb{R} \; \forall x, y \in \mathbb{R}^n \; B(x, cy) = cb(x, y)$

Moreover,

- The bilinear form B is said to be **symmetric** if

$$\forall x, y \in \mathbb{R}^n \quad B(x, y) = B(y, x).$$

- The bilinear form B is said to be **skew-symmetric** if

$$\forall x, y \in \mathbb{R}^n \quad B(x, y) = -B(y, x).$$

- The bilinear form B is said to be **non-degenerate** if the following statement is true

$$B(x, y) = 0 \;\; \forall y \in \mathbb{R}^n \implies x = 0.$$

- A symmetric bilinear form B is said to be **positive definite** if

$$B(x, x) > 0 \;\; \forall x \neq 0.$$

Example (Standard Inner Product on $\mathbb{R}^n$). The standard inner product in $\mathbb{R}^n$

$$\langle x, y\rangle = \sum_{i=1}^{n} x_i y_i$$

is a symmetric positive definite bilinear form on $\mathbb{R}^n$.

For example, let's verify condition (1) for bilinearity:

$$\begin{aligned}\langle x+y, z\rangle &= \sum_{i=1}^{n}[(x_i+y_i)z_i] \\ &= \sum_{i=1}^{n}[x_i z_i + y_i z_i] \\ &= \sum_{i=1}^{n} x_i z_i + \sum_{i=1}^{n} y_i z_i \\ &= \langle x, z\rangle + \langle y, z\rangle.\end{aligned}$$

I leave you to check the other conditions yourself.

In general, a symmetric positive definite bilinear form on a real vector space V, by definition, is called an **inner product** on V.

Example (Standard Symplectic Form on $\mathbb{R}^n$). Define $w + \mathbb{R}^{2n} \times \mathbb{R}^{2n} \to \mathbb{R}$ as follows:

$$\forall x, y \in \mathbb{R}^{2n} \quad w(x,y) = \sum_{j=1}^{n} x_j y_{n+j} - x_{n+j} y_j.$$

For example,

$$\begin{aligned} n = 1 &\Longrightarrow w(x,y) = x_1 y_2 + x_2 y_1 \\ n = 2 &\Longrightarrow w(x,y) = (x_1 y_3 - x_3 y_1) + (x_2 y_4 - x_4 y_2) \\ &\vdots \end{aligned}$$

If we let $\Omega = \begin{pmatrix} O_{n\times n} & I_{n\times n} \\ -I_{n\times n} & O_{n\times n}\end{pmatrix}$, then we can write the standard symplectic form in the following way:

$$w(x,y) = \langle x, \Omega y\rangle$$

where $\langle , \rangle$ is the standard inner product in $\mathbb{R}^{2n}$.

Proposition (Preserving the standard symplectic form on $\mathbb{R}^{2n}$). Suppose $A \in M(2n, \mathbb{R})$. We say that A preserves the standard symplectic form on $\mathbb{R}^{2n}$ if and only if $A^T \Omega A = \Omega$.

Proof. We say that $A \in M(2n, \mathbb{R})$ preserves the standard symplectic form on $\mathbb{R}^{2n}$ if and only if

$$\begin{aligned}\forall x, y \in \mathbb{R}^{2n} \quad w(Ax, Ay) = w(x,y) &\Longleftrightarrow \langle Ax, \Omega(Ay)\rangle = \langle x, \Omega y\rangle \\ &\Longleftrightarrow \langle x, A^T \Omega A y\rangle = \langle x, \Omega y\rangle \\ &\Longleftrightarrow \forall y \in \mathbb{R}^{2n} A^T \Omega A y = \Omega y \\ &\Longleftrightarrow A^T \Omega A = \Omega.\end{aligned}$$

■

Proposition. If $A^T\Omega A = \Omega$, then A is invertible.

Proof. Suppose $A^T\Omega A = \Omega$, then we have

$$\begin{aligned}
&\det(A^T\Omega)A = \det(A) \\
&\implies \det(A^T)\det(\Omega)\det(A) = \det(\Omega) \\
&\implies (\det(A))^2(\det(\Omega)) = \det(\Omega) \\
&\implies (\det(A))^2 = 1 \\
&\implies \det(A) = \pm 1 \\
&\implies A \text{ is invertible.}
\end{aligned}$$

■

Proposition. $A^T\Omega A = \Omega$ if and only if $-\Omega A^T\Omega = A^{-1}$ where A is invertible.

Proof. We have

$$\begin{aligned}
-\Omega A^T\Omega = A^{-1} &\underbrace{\iff}_{\text{left multiply by } \Omega^{-1}} -A^T\Omega = \Omega^{-1}A^{-1} \\
&\underbrace{\iff}_{\text{right multiply by } A} -A^T\Omega A = \Omega^{-1} \\
&\underbrace{\iff}_{\Omega^{-1}=-\Omega} -A^T\Omega A = \Omega \\
&\iff A^T\Omega A = \Omega.
\end{aligned}$$

■

Example (Real Symplectic Group)**.** $\mathrm{Sp}(2n,\mathbb{R}) = \{A \in M(2n,\mathbb{R}) : A^T\Omega A = \Omega\}$ is a subgroup of $\mathrm{GL}(2n,\mathbb{R})$.

Proof. We just proved that every matrix in $\mathrm{SP}(2n,\mathbb{R})$ is invertible. So,

$$\mathrm{SP}(2n,\mathbb{R}) \subseteq \mathrm{GL}(2n,\mathbb{R}).$$

- We prove that $\mathrm{SP}(2n,\mathbb{R})$ is closed under multiplication. Suppose $A, B \in \mathrm{SP}(2n,\mathbb{R})$. We have

$$\begin{aligned}
(AB)^T\Omega(AB) &= B^T(A^T\Omega A)B \\
&= B^T\Omega B \\
&= \Omega.
\end{aligned}$$

  Hence, we have $AB \in \mathrm{SP}(2n,\mathbb{R})$.

- $\mathrm{SP}(2n,\mathbb{R})$ contains the inverse of each of its elements. Suppose $A \in \mathrm{SP}(2n,\mathbb{R})$. We have

$$\begin{aligned} A^T\Omega A = A &\implies (A^T)^{-1}\Omega A = (A^T)^{-1}\Omega \\ &\implies \Omega = (A^T)^{-1}\Omega A^{-1} \\ &\implies \Omega = (A^{-1})^T\Omega A^{-1} \\ &\implies A^{-1} \in \mathrm{SP}(2n,\mathbb{R}) \end{aligned}$$

From the above, we can see that $\mathrm{SP}(2n,\mathbb{R})$ is a subgroup of $\mathrm{GL}(2n,\mathbb{R})$. Now, we will prove that $\mathrm{SP}(2n,\mathbb{R})$ is a Lie subgroup of $\mathrm{GL}(2n,\mathbb{R})$. Define the map

$$G : \mathrm{GL}(2n,\mathbb{R}) \subseteq \mathbb{R}^{4n^2} \to M(2n,\mathbb{R}) \cong \mathbb{R}^{4n^2} \text{ by } A \mapsto A^T\Omega A - \Omega.$$

Clearly, $\mathrm{SP}(2n,\mathbb{R}) = F^{-1}(\{0\})$. It can be shown that

$$\forall A \in \mathrm{GL}(2n,\mathbb{R}) \ \ DF(A) \text{ has rank } \frac{4n^2-2n}{2} = 2n^2 - n.$$

Hence, $\mathrm{SP}(2n,\mathbb{R})$ is an embedded submanifold of $\mathrm{GL}(2n,\mathbb{R})$ with dimension $4n^2-(2n^2-n) = 2n^2+n$. Thus, $\mathrm{SP}(2n,\mathbb{R})$ is a Lie subgroup of $\mathrm{GL}(2n,\mathbb{R})$. ■

Definition (Homomorphisms, Isomorphisms, Automorphisms)**.** Let G and H be two groups.

- A function $\varphi : G \to H$ is said to be a **homomorphism** if it preserves group multiplication

$$\forall g_1, g_2 \in G \ \ \varphi(g_1g_2) = \varphi(g_1)\varphi(g_2).$$

- A bijective homomorphism is called an **isomorphism**. If there is an Isomorphism between G and G, we say G and H are isomorphic, and we write $G \cong H$.
- An **endomorphism** of G is a homomorphism from G to itself.
- An **automorphism** of G is an isomorphism from G to itself.

Definition (Lie Group Homomorphism)**.** Let G and H be two Lie groups.

- A function $f : G \to H$ is said to be a **Lie group homomorphism** if
  (i) f is a group homomorphism
  (ii) f is smooth
- A function $f : G \to H$ is said to be a **Lie group isomorphism** if
  (i) f is a group isomorphism
  (ii) f is smooth
  (iii) f^{-1} is smooth

  If f is smooth and its inverse is smooth, then f is a diffeomorphism.

Example (The Determinant is a Lie Group Homomorphism)**.** The determinant $\det : \mathrm{GL}(n,\mathbb{R}) \to \mathbb{R}^*$ is a Lie Group Homomorphism:

(i) $\forall A, B \in \mathrm{GL}(n,\mathbb{R})$, we have $\det(AB) = \det(A)\det(B)$. Hence, det is a group homomorphism.

(ii) $\det A$ is a polynomial in terms of the entries of A and so differentiable. Hence, the det is smooth.

Similarly, the det$GL(n, \mathbb{C}) \to \mathbb{C}^*$ is a Lie Group Homomorphism.

Example (Left Translation/right translation)**.** Let G be a Lie Group and let $g \in G$. The **left translation by** g is the map

$$L_g : G \to G \;\; h \mapsto gh.$$

The **right translation by** g is the map

$$R_g : G \to G \;\; h \mapsto hg.$$

Notice that L_g is a composition of smooth maps and so it is smooth:

$$G \to G \times G \to G \;\; h \mapsto (g, h) \mapsto gh.$$

Starting from left to right, we see that the first mapping is smooth because

- Fact 1: $F = (F_1, F_2)$ is smooth if and only if F_1 and F_2 are smooth.
- Fact 2: $h \mapsto g$ is a constant function implies that it is smooth.
- Fact 3: $h \mapsto h$ is the identity which is also smooth.

The second mapping $G \times G \to G$ is smooth because multiplication in G is smooth since G is a Lie group.

In fact, $L_g : G \to G$ is a diffeomorphism because the inverse L_g is $L_{g^{-1}}$:

$$\forall h \;\; L_{g^{-1}} \circ L_g(h) = g^{-1}(gh) = h.$$

Unfortunately, L_g in general is NOT a group homomorphism; that is, in general

$$L_g(h_1 h_2) \neq L_g(h_1) L_g(h_2).$$

Similarly, the right translation map also satisfies the properties above but is NOT a group homomorphism.

Example (Conjugation by $g \in G$)**.** Let G be a Lie group and let $g \in G$. The conjugation by g is the map:

$$C_g : G \to G \;\; h \mapsto ghg^{-1}.$$

Note that C_g is a diffeomorphism because it is a composition of diffeomorphisms:

$$C_g = L_g \circ R_{g^{-1}}.$$

In fact, $C_g^{-1} = C_{g^{-1}}$. Also, note that

$$\forall h_1, h_2 \in G \;\; C_g(h_1 h_2) = g(h_1 h_2)g^{-1} = (gh_1 g^{-1})(gh_2 g^{-1}) = C_g(h_1) C_g(h_2).$$

Hence, $C_g : G \to G$ is a Lie group isomorphism (i.e a Lie group automorphism).

3 Week 7

3.1 Kernel and Image

Definition (Kernel and Image)**.** Let G and H be two groups. Let $f : G \to H$ be a homomorphism.

- The **kernel** of f is the set

$$\ker f = \{g \in G : f(g) = 1_H\} = f^{-1}(\{1_H\}) \subseteq G.$$

- The **image** of f is
$$\mathrm{Im}f = \{f(g) : g \in G\} = f(G) \subseteq H.$$

Both of these sets are subgroups of G and H, respectively.

Example. Consider the determinant $\det : \mathrm{GL}(n, \mathbb{R}) \to \mathbb{R}^*$ is a homomorphism. Thus,
$$\ker(\det) = \{A \in \mathrm{GL}(n, \mathbb{R}) : \det A = 1\} = \mathrm{SL}(n, \mathbb{R}).$$

3.2 A Common Theme in Mathematics

Remark (Topology)**.** If X and Y are two topological spaces, we may consider them to be the same if there exists a function $f : X \to Y$ such that

(1) f is bijective

(2) f is continuous

(3) f^{-1} is continuous

In the context of manifold theory, if M and N are two manifolds, we may consider them to be the same if there exists $f : M \to N$ such that

(1) f is bijective

(2) f is smooth

(3) f^{-1} is smooth

In the context of group theory, if G and H are two groups, we may consider them to be the same if

(1) f is bijective

(2) f is a homomorphism

The last remark gives us an explanation of why (ii) is important.

3.3 Kernel of a Lie Group Homomorphism

Remark (Fact from Linear Algebra)**.** Suppose A and B are two $n \times n$ matrices and A is invertible. Then
$$\mathrm{rank}(AB) = \mathrm{rank}(B) \quad \mathrm{rank}(BA) = \mathrm{rank}(B).$$

Theorem (Kernel of a Lie Group Homomorphism)**.** Let G and H be Lie groups. Let $f : G \to H$ be a Lie group homomorphism. Then

(i) The rank of the derivative of f is constant for all points in G; that is, there exists $k \in \mathbb{N}$ such that for all $x \in G$, $\mathrm{rank}(Df(x)) = k$.

(ii) $\ker(f)$ is an embedded submanifold of G of dimension $\dim(G) - k$ (and so it is a Lie subgroup of G)

(iii) $\mathrm{Im}(f)$ can be equipped with a smooth structure that makes it a Lie subgroup of H.

Proof. Note that (ii) is a direct consequence of (i) and Fact 2 from many weeks ago. Here we will prove (i). It suffices to show that for all $x \in G$, $\mathrm{rank}(Df(x)) = \mathrm{rank}(Df(e))$ where $e = 1_G$. For all

$a, x \in G$, we have

$$f(x) = f(aa^{-1}x) = f(a) \cdot f(a^{-1}x)$$
$$= (L_{f(a)} \circ f \circ L_{a^{-1}})(x).$$

By the chain rule,

$$Df(x) = DL_{f(a)}(f(a^{-1}x)) \cdot Df(a^{-1}x) \cdot DL_{a^{-1}}(x).$$

Since $L_{f(a)}$ and $L_{a^{-1}}$ are diffeomorphisms, $DL_{f(a)}$ and $DL_{a^{-1}}$ are invertible at any point. From a fact in linear algebra, for all $x, a \in G$

$$\text{rank}(Df(x)) = \text{rank}(Df(a^{-1}x)).$$

In particular, if we set $a = x$, we get

$$\text{rank}(Df(x)) = \text{rank}(Df(e))$$

which is our desired result. ■

Remark. Suppose G is a diffeomorphism. We have

$$(G^{-1} \circ G)(x) = x \Longrightarrow DG^{-1}(G(x))DG(x) = I$$

Hence, $DG(x)$ is invertible.

3.4 Subgroups and Cosets

Definition (Subgroups and cosets (very important!))**.** Let G be a group. Given a subgroup H of G and $g \in G$, the **left coset** of H determined by g is the set

$$gH = \{gh : h \in H\}.$$

The **right coset** of H determined by g is the set

$$Hg = \{hg : h \in H\}.$$

Note that H itself is a coset for $g = 1_G$.

You need a subgroup and an element to talk about a coset.

Example ($G = (\mathbb{R}^2, +), H =$ y-axis (Very important!))**.** Observe that $H = \{(0, y) : y \in \mathbb{R}\}$ is a subgroup of $G = (\mathbb{R}^2, +)$. Let's determine the left cosets of H. For all $(a, b) \in \mathbb{R}^2$, we have

$$(a, b) + H = \{(a, b) + (0, y) : y \in \mathbb{R}\} = \{(a, 0 + y) : y \in \mathbb{R}\}$$
$$= \{(a, y) : y \in \mathbb{R}\}.$$

This represents the vertical line at $x = a$. So, in this example, G decomposes into disjoint cosets. Roughly speaking, the plane is the union of parallel lines. This organizes the mess that is $\mathbb{R}^2$ into more manageable chunks to understand. Is this by accident? NO!

Note that two distinct $g \in G$ can give us the same coset (or vertical line in our example). That is, consider the equivalent cosets

$$(1, 0) + H = (1, 1) + H = \text{ vertical line at } x = a.$$

Theorem. Let G be a group and H be a subgroup of G.

(i) $G = \bigcup_{g \in G} gH$

(ii) Suppose $g_1, g_2 \in G$. Then either $g_1H = g_2H$ or $g_1H \cap g_2H = \emptyset$.

(iii) $g_1H = g_2H$ if and only if $g_1^{-1}g_2 \in H$.

Proof. (i) For all $g \in G$, clearly $gH \subseteq G$ since gH is a subgroup of G. Therefore,

$$\bigcup_{g \in G} gH \subseteq G.$$

Suppose $g' \in G$. Since $1 \in H$, we have $g'1 \in g'H$. Thus, $g' \in g'H \subseteq \bigcup_{g \in G} gH$. So, $G \subseteq \bigcup_{g \in G} gH$. Hence,

$$G = \bigcup_{g \in G} gH.$$

(ii) It suffices to show that if $g_1H \cap g_2H \neq \emptyset$, then $g_1H = g_2H$. Suppose $g \in g_1H \cap g_2H$. Thus,

$$g = g_1h_1 = g_2h_2 \text{ for some } g_1 \text{ and } g_2 \text{ in } H.$$

We have

$$\begin{aligned} g_1 = g_2h_2h_1^{-1} &\implies g_1H = g_2h_2h_1^{-1}H \\ &\implies g_1H = g_2(h_2h_1^{-1}H) \\ &\implies g_1H = g_2H. \end{aligned}$$

(iii) ($\impliedby$) Suppose $g_1^{-1}g_2 \in H$. Our goal is to show that $g_1H = g_2H$. By (ii), it suffices to show that $g_1H \cap g_2H \neq \emptyset$. We have

$$\begin{aligned} g_1^{-1}g_2 \in H &\implies g_1(g_1^{-1}g_2) \in g_1H \\ &\implies g_2 \in g_1H. \end{aligned}$$

Similarly, we have $g_2 \in g_2H$. Thus, $g_2 \in g_1H \cap g_2H$. So, we have $g_1H \cap g_2H \neq \emptyset$.

($\implies$) Assume $g_1H = g_2H$. We claim that $g_1^{-1}g_2 \in H$.

$$\begin{aligned} g_1H = g_2H &\implies g_2 \in g_1H \\ &\implies g_2 = g_1h \ \text{ for some } h \in H \\ &\implies g_1^{-1}g_2 = h \ \text{ for some } h \in H \\ &\implies g_1^{-1}g_2 \in H. \end{aligned}$$

■

Theorem (Congruence modulo H)**.** Let G be a group and H be a subgroup. Define the relation **congruence modulo** H on G as follows:

$$g_1 \sim g_2 \iff g_1^{-1}g_2 \in H.$$

Then the above relation is an equivalence relation , and its equivalence classes are precisely the left cosets of H.

Remark.
- When we look at a group G, one very natural perspective is to view it as a disjoint union of cosets of a subgroup H.
- This viewpoint organizes the elements of G and often makes it easier to extract information about the structure of the group.
- In a way, the collection of cosets is simpler and more manageable than the group itself.

Remark.
- It is tempting to try to make this collection of cosets into a group, using the natural operation
$$(g_1H)(g_2H) := (g_1g_2)H.$$
However, this does not always work; in general, the operation may be ill-defined.
- The question is as follows: *Under what conditions is the above operation well-defined?*

Lemma. Let G be a group and H be a subgroup of G. The following statements are equivalent:

(1) $\forall g \in G$, $\forall h \in H$, $ghg^{-1} \in H$

(2) $\forall g \in G$, $\forall h \in H$, $gHg^{-1} = H$

(3) $\forall g \in G$, $gH = Hg$.

3.5 Normal Subgroups

Definition (Normal Subgroups)**.** In the case that any of the above equivalent conditions holds, we say H is a normal subgroup of G.

Notationally, normal subgroups can be represented by $H \trianglelefteq G$. If G is abelian, any subgroup of G is normal.

Theorem. Suppose G is a group and H is a subgroup of G. The operation
$$(g_1H) \cdot (g_2H) = (g_1g_2) \cdot H$$
is well-defined on the collection of left cosets of H if and only if H is a normal subgroup of G.

Proof. ($\Longrightarrow$) Assume that the operation above is well-defined. We want to show that H is a normal subgroup of G. That is, it suffices to show that for all $g \in G$ and $h \in H$, we have
$$ghg^{-1} \in H.$$
That is, we want to show that (using the previous result $aH = bH \Longleftrightarrow b^{-1}a \in H$)
$$g^{-1}H = (gh)^{-1}H.$$
That is, we want to show that
$$g^{-1}H = h^{-1}g^{-1}H \tag{*}$$

Since the coset operation is well defined, we have

$$\begin{aligned} g^{-1}H = (1g^{-1})H = (1H)(g^{-1}H) &\underbrace{=}_{1H=H=h^{-1}H} (h^{-1}H)(g^{-1}H) \\ &= (h^{-1}g^{-1})H. \end{aligned}$$

($\Longleftarrow$) Assume that $H \trianglelefteq G$. We want to show that the above operation is well-defined. Suppose $g_1 H = g_1' H$ and $g_2 H = g_2' H$. Our goal is to show that

$$(g_1 g_2)H = (g_1' g_2')H. \tag{*}$$

Since H is normal, (that is, $g_2' H = H g_2'$ and $g_1' H = H g_1'$)

$$\begin{aligned} g_1' g_2' H = g_1' H g_2' &= g_1 H g_2' && (g_1' H = g_1 H) \\ &= g_1 g_2' H && (H \trianglelefteq G \Longrightarrow H g_2' = g_2' H) \\ &= g_1 g_2 H. && (g_2' H = g_2 H) \end{aligned}$$

and we are done. ■

Theorem (Quotient Theorem for Groups)**.** If H is a normal subgroup of G, then the operation

$$(g_1 H) \cdot (g_2 H) = (g_1 h_2)H$$

is well-defined on cosets

The group of cosets is called the **quotient group of** G **by** H, and is written G/H.

Remark. The map $\pi : G \to G/H$ where $g \mapsto gH$ is a surjective homomorphism and is called the **quotient map**.

By studying quotient groups of a group which, in a sense, organizes the group's elements into simpler chunks, we can gain valuable information about the group itself. For example, we can show that if $G/Z(G)$ is cyclic, then G is abelian (where $Z(G)$ is the center of G).

The kernel of the quotient map will be a normal subgroup of G.

3.6 Making G/N a Lie Group

Remark (Observation)**.** Suppose G is a Lie group and N is a normal algebraic subgroup of G. From this we can construct a new algebraic group G/N. In the context of Lie Theory, an important question arises:

Can we make G/N a Lie group?

It seems reasonable to look for a smooth structure on G/N that at least ensures that the quotient map $\pi : G \to G/N$ is smooth.

What is the kernel of π? It is

$$\begin{aligned} \ker(\pi) = \pi^{-1}\{1_G N\} &= \{g \in G : \pi(g) = 1_G N\} \\ &= \{g \in G : \pi(g) = N\} \\ &= \{g \in G : gN = N\} \\ &= N. \end{aligned}$$

(**very important!**)

What can we conclude from this result? If we manage to find a smooth structure on G/N that ensures $\pi : G \to G/N$ is smooth, then N must be necessarily a topologically closed subset of G (recall that if π

is smooth, then it is continuous and preimages of continuous functions of closed sets are closed). If it is not closed, then there is no way to define a smooth structure on the homomorphism π.

Theorem (Quotient Theorem for Lie Groups). Let G be a Lie group, $N \trianglelefteq G$, and N is topologically closed subset of G. Then G/N can be equipped with a smooth structure that turns it into a Lie group such that the quotient map $\pi : G \to G/N$ is a surjective smooth map.

Example ($\mathbb{R}/\mathbb{Z}$). Note that $G = (\mathbb{R}, +)$ is a Lie group and $N = (\mathbb{Z}, +)$ is a subgroup of G. Since G is abelian, $N \trianglelefteq G$. Now, is N a closed subset? Yes, because the complement of $\mathbb{Z}$ (that is $\mathbb{R} \setminus \mathbb{Z}$) is open, so $\mathbb{Z}$ is a closed subset of $\mathbb{R}$. Using the Quotient theorem for Lie groups, we have that $\mathbb{R} \setminus \mathbb{Z}$ is a Lie group.

As a matter of fact, the mapping

$$\begin{aligned} F : \mathbb{R}/\mathbb{Z} &\longrightarrow S^1 = \mathrm{U}(1) \\ x + \mathbb{Z} &\longmapsto e^{2\pi i x} \end{aligned}$$

is a well-defined isomorphism of Lie group. And so,

$$\mathbb{R}/\mathbb{Z} \cong S^1.$$

Why is this well-defined? Suppose $x + \mathbb{Z} = y + \mathbb{Z}$. We need to show that $e^{2\pi i x} = e^{2\pi i y}$. We have

$$\begin{aligned} x + \mathbb{Z} = y + \mathbb{Z} &\Longrightarrow x - y \in \mathbb{Z} \\ &\Longrightarrow e^{2\pi i (x-y)} = \cos(2\pi(x-y)) + i \sin(2\pi(x-y)) \\ &\Longrightarrow e^{2\pi i (x-y)} = 1 + i0 = 1 \\ &\Longrightarrow e^{2\pi i x} \cdot e^{-2\pi i y} = 1 \\ &\Longrightarrow e^{2\pi i x} = e^{2\pi i y}. \end{aligned}$$

Similarly, we can show that

$$\mathbb{R}^n/\mathbb{Z}^n \cong (S^1) \times \cdots \times (S^1) \qquad (n\text{-dimensional torus})$$

mapped by $(x_1, \ldots, x_n) + \mathbb{Z}^n \mapsto (e^{2\pi i x_1}, e^{2\pi i x_2}, \ldots, e^{2\pi i x_n})$.

Example. Let $G = \mathrm{SU}(2)$ be a Lie group, $N = \{-I, +I\}$ is a normal subgroup of $\mathrm{SU}(2)$, then

$$\forall g \in \mathrm{SU}(2) \quad g(\pm I)g^{-1} = \pm g g^{-1} = \pm I \in N$$

where N is closed G (N is finite subset of G). Hence, $\mathrm{SU}(2)/\{-I, I\}$ is a Lie group. In fact,

$$\mathrm{SU}(2)/\{-I, +I\} \cong \mathrm{SO}(3).$$

4 Week 8

4.1 Conjugate subgroups, Conjugacy Class

Let G be a group. Recall that for each $g \in G$, the map conjugation by g, $C_g : G \to Gh \mapsto ghg^{-1}$ is a homomorphism. Since $C_g : G \to G$ is a homomorphism, for any subgroup, $H \leq G$, $C_g(H)$ will be a subgroup of G. Note that if H is a normal subgroup of G, then

$$C_g(H) = gHg^{-1} = H. \ \forall g \in G$$

So,

$$\{C_g(H) : g \in G\} = H$$

which is just H. We can use this set as a measure of how our subgroup deviates from being normal. This motivates the following definition.

Definition (Conjugacy Class). For any subgroup $H \leq G$, we define the **conjugacy class** of H as follows:

$$\{C_g(H) : g \in G\} = \{gHg^{-1} : g \in G\}.$$

If $\tilde{H} \in \{C_g(H) : g \in G\}$, we say that subgroups H and $\tilde{H}$ are conjugate.

Example. Let $\theta \in \mathbb{R}$ be a fixed number. Consider the cyclic subgroup of SO(3) generated by R^z_θ:

$$H = \langle R^z_\theta \rangle = \{R^z_{k\theta} : k \in \mathbb{Z}\}.$$

Note that for all $g \in \mathrm{SO}(3)$, we have

$$\begin{aligned} gHg^{-1} &= \{gR^z_{k\theta}g^{-1} : k \in \mathbb{Z}\} \\ &= \text{the cyclic subgroup of SO(3)} \\ &\quad \text{generated by rotation } \theta \text{ radians about the axis} \\ &\quad \text{determined by the 3rd column of } g. \end{aligned}$$

Hence, we have

$$\begin{aligned} \text{Conjugacy class of } H &= \{gHg^{-1} : g \in \mathrm{SO}(3)\} \\ &= \text{the collection of all cyclic subgroups} \\ &\quad \text{of SO(3) generated by rotation } \theta \text{ radians about} \\ &\quad \text{an arbitrary fixed axis.} \end{aligned}$$

Theorem (First Isomorphism Theorem). Let $f : G \to H$ be a group homomorphism. Then $\varphi : G/\ker(f) \to \mathrm{Im}(f)$ where $g\mathrm{Ker}(f) \mapsto f(g)$.

Example (Observation of rotational matrices).

- Rotation by an angle θ about the origin in the xy-plane can be described by the matrix

$$\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}.$$

- Rotation by an angle θ about the z-axis in the xyz-space can be described by the matrix

$$R^z_\theta = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Example. Let ℓ be a directed line that passes through the origin in the xyz-space. In what follows, we will find a formula for rotation by an angle θ about ℓ (an arbitrary axis). Let $e_1 = (1,0,0), e_2 = (0,1,0), e_3 = (0,0,1)$ be the standard basis. Now, choose a new orthonormal basis $\{\hat{e_1}, \hat{e_2}, \hat{e_3}\}$ such that

(1) It is right-handed ($\hat{e}_3 = \hat{e}_1 \times \hat{e}_2$)

(2) $\hat{e}_3$ lies on ℓ (and points in the same direction as ℓ). Let $A = P_{\beta \leftarrow \hat{B}}$ be the change of basis matrix from $\hat{\beta}$ coordinate to β coordinates. Suppose w is a vector in $\mathbb{R}^3$. We can write

$w = w_1e_1 + w_2e_2 + w_3e_3$ and $w = \hat{w_1}\hat{e_1} + \hat{w_2}\hat{e_2} + \hat{w_3}\hat{e_3}$. Recall that

$$[w]_\beta = \begin{pmatrix} w_1 \\ w_2 \\ w_3 \end{pmatrix}, \quad [w]_{\hat{\beta}} = \begin{pmatrix} \hat{w_1} \\ \hat{w_2} \\ \hat{w_3} \end{pmatrix}.$$

4.2 Review of Linear Algebra

Suppose V and W are vector spaces with dimensions n and m, respectively. Suppose T is a linear transformation from V to W.

Remark (Huge Fact). If we choose a basis $\{b_1, \dots, b_n\}$ for V a basis $\{f_1, \dots, f_m\}$ for W, then T can be represented by an $m \times n$ matrix, denoted by $M(T)$ in the following sense: Given any vector $v = v_1b_1 + \cdots + v_nb_n$ is equal to $w_1f_1 + w_2f_2 + \cdots + w_mf_m$ where

$$\begin{pmatrix} w_1 \\ \vdots \\ w_m \end{pmatrix} = M(T) \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix}.$$

It can be shown that

$$M(T) = \begin{pmatrix} T(b_1) & T(b_2) & \cdots & T(b_n) \end{pmatrix}$$

where for each $1 \le i \le n$, $T(b_i)$ will be expressed in terms of the basis in $\{b_1, \dots, b_n\}$.

Now, suppose V is a vector space and $\{b_1, \dots, b_n\}$ is one basis for V and $\{\hat{b_1}, \dots, \hat{b_n}\}$ is a second basis for V. We may consider the identity map

$$I : V \to V \quad I(x) = x$$

and equip the domain with $\{\hat{b_1}, \dots, \hat{b_n}\}$ basis, and equip the codomain with $b_1, \dots, b_n$. Then the $M(I)$ is exactly the change of coordinate matrix $P_{\beta \leftarrow \hat{\beta}}$ from $\hat{\beta}$ to β. Note that $A \in \mathrm{SO}(3)$ because the columns of A are orthonormal and A preserves orientation. Let $v = v_1e_1 + v_2e_2 + v_3e_3$ be a vector in $\mathbb{R}^3$; that is, $[v]_\beta = (v_1, v_2, v_3)$. Let w be the vector that is obtained by rotating v by θ radians about the ℓ axis. We claim that

$$[w]_\beta = AR^z_\theta A^{-1}[v]_\beta.$$

Indeed, $A^{-1}[v]_\beta = [v]_{\hat{\beta}}$. Now, in the $\hat{\beta}$ basis, the ℓ axis is the same as the z-axis. Thus, the vector w that is obtained by rotation, θ radians about ℓ will have coordinates given by

$$[w]_{\hat{\beta}} = R^z_\theta[v]_{\hat{\beta}} = R^z_\theta A^{-1}[v]_\beta.$$

Therefore,

$$[w]_\beta = A[w]_\beta = AR^\theta_z A^{-1}[v]_\beta.$$

That is,

$$[w]_\beta = (AR^z_\theta A^{-1})[v]_\beta.$$

4.3 Remark about First Isomorphism Theorem

Suppose G is a group and K is a normal subgroup of G. Then, as we know, the quotient map $\pi : G \to G/K$ where $g \mapsto gk$ is a surjective homomorphism. The above theorem for groups tells us that any surjective homomorphism with domain G is of the above form. That is, if $f : G \to H$ is a surjective homomorphism, we have

$$H \cong G/\mathrm{Ker}(f).$$

Hence, your surjective homomorphism is, indeed, $G \to G/\mathrm{Ker}(f)$. This theorem is giving us a way to construct all homomorphisms with domain G. In fact, we can make a connection to the following remarkable result from Category Theory.

4.4 Consequence of Yoneda Lemma

If you want to understand a mathematical object X, it suffices to understand all the structure preserving maps with domain X.

4.5 Second Isomorphism Theorem

Let G be a group. Let $K \trianglelefteq G$. As we know, the quotient map $\pi : G \to G/K$ is a surjective homomorphism. Suppose H is a subgroup of G. What is exactly $\pi(H)$ (we know that it must be a subgroup of G/K)? By the correspondence theorem, we know that

$$\pi(H) = (\text{a subgroup of } G \text{ that contains } K)/K.$$

The answer is:

$$\pi(H) = HK/K.$$

Theorem (Second Isomorphism Theorem)**.** Let G be a group, H is a subgroup of G, K is a normal subgroup of G, then

$$\varphi : HK/K \to H/(H \cap K), \quad hk \mapsto h(H \cap K)$$

is an isomorphism

4.6 Third Isomorphism Theorem

Let G be a group, according to the first isomorphism theorem, in order to study homomorphisms with domain G, it suffices to study quotients G/N where N is a normal subgroup of G. Will it be helpful now to consider the quotients of G/N? No! There will be no new information if you construct the quotients of G/N.

Theorem (Third Isomorphism Theorem)**.** Let G be a group, K be a normal subgroup of G, N is a subgroup of K, and N is a normal subgroup of G. Then

$$\varphi : (G/N)/(K/N) \to G/K, \quad gN(K/N) \mapsto gK$$

is an isomorphism.

4.7 The Correspondence Theorem

Let G be a group and K be a normal subgroup of G. There are two questions we need to answer.

(1) What are the subgroups of G/K?

(2) What are the normal subgroups of G/K?

The answer to these questions are the following:

(1) There is a one-to-one correspondence between the set of all subgroups of G that contain K and the set of all subgroups of G/K:

- (i) If $H \leq G$ and $R \leq G/K$, then $H/K \leq G/K$.
- (ii) If $R \leq G/K$, then $R = H/K$ for some $H \leq G$ and $K \subseteq H$.
- (ii) The above correspondence preserves normality:

$$H/K \trianglelefteq G/K \iff H \trianglelefteq G, K \subseteq H.$$

4.8 Simple Groups

Definition (Simple Groups)**.** Let G be an algebraic group. We say that G is **simple** if

(1) $G \neq \{1_G\}$

(2) The only normal subgroups of G are G itself and $\{1_G\}$.

Cannot make the group much simpler; that is, we cannot quotient the group out by a normal subgroup even further.

Remark (Observation)**.** Many-to-one functions help us simplify/organize the domain of the function. An example is the quotient map $\pi : G \to G/N$ defined by $g \mapsto gN$. This organizes the domain of π into cosets.

Remark. The following two statements are equivalent:

(1) G is not simple

(2) There exists a many-to-one homomorphism of G onto a group G' other than the homomorphism sending all elements to 1.

That is, G' can be viewed as a simpler version of G (or, at any rate, not more complicated than G).

Remark.

- A **simple Lie group** is not usually defined as a Lie group that is simple in the sense of abstract group theory.
- The most common definition is that a Lie group G is called **simple** if its corresponding Lie algebra is a simple Lie algebra.
- As we shall see, with this definition, a simple Lie group may not be simple as an abstract group.

4.9 Direct Product of Groups

Definition (Direct Product of Groups)**.** If G_1 and G_2 are two groups, the **direct product** of G_1 and G_2 is the set $G_1 \times G_2$ with the group structure defined by the following multiplication rule:

$$(g_1, g_2) \cdot (g_1', g_2') = (g_1 g_1', g_2 g_2')$$

and the identity is

$$1_{G_1 \times G_2} = (1_{G_1}, 1_{G_2}).$$

If G_1 and G_2 are Lie groups, then $G_1 \times G_2$ is a Lie group.

(1) $G_1 \times G_2$ is an algebraic group (previous slide)

(2) G_1 is a smooth manifold and G_2 is a smooth manifold, from a month ago, we have $G_1 \times G_2$ is a smooth manifold.

(3) multiplication map: $(g_1, g_2) \cdot (g_1', g_2') = (g_1 g_1', g_2 g_2')$. The multiplication in the first and second components are smooth operations. Hence, the multiplication in the direct product is also smooth.

(4) Inversion: $(g_1, g_2) \mapsto (g_1^{-1}, g_2^{-1})$.

Example. We know that $S^1 = U(1) = \mathrm{SO}(2)$ is a Lie group. It is, indeed, a compact and connected.

Thus, $\underbrace{S^1 \times \cdots S^1}_{k \text{ times}}$ is also a Lie group. Note that this set is

(1) connected

(2) compact

(3) abelian

Why is S^1 closed? We know that $S^1 = f^{-1}(1)$ where $f : \mathbb{R}^2 \to \mathbb{R}$, $f(x, y) = x^2 + y^2$ and that because f is continuous, we know that the preimage of the closed set $\{1\}$ is closed. Clearly, we also see that S^1 is bounded. Since each set is compact, and there are k number of them, we see that because there are a finite product of compact sets, we know that the finite product must also be compact. Note that it is also true that a finite product of connected topological spaces is connected.

In fact, the only k-dimensional, where $k \geq 1$, Lie group that is compact, connected, and abelian is T^k. That is, if G is a k-dimensional compact connected abelian Lie group, then $G \cong T^k$ (where $\cong$ is a Lie group homomorphism). Here are some core ideas of the proof:

(1) Use the fact G is abelian to show that the Lie Algebra of G is isomorphic to $\mathbb{R}^k$ with $[x, y] = 0$.

(2) Use the facts that G is compact and connected to show that

$$\exp : g \to G \text{ is surjective.}$$

(3) Use the fact that G is abelian to show that $\exp : g \to G$ is indeed a group homomorphism; that is, $g/\ker(\exp \cong G$

(4) Use the fact that G is compact and that $g/\ker(\exp) \cong \mathbb{Z}^k$.

(5) Conclude that

$$G \cong g/\ker(\exp) \cong \mathbb{R}^k/\mathbb{Z}^k \cong (S^1)^k = T^k.$$

End of Week 8

5 Week 9

In mathematics it is often helpful, when studying an object X (such as an algebraic structure, a metric space, or a normed space) to identify certain subsets/subobjects/substructures $E \subseteq X$ that are easier to work with or posses nice properties.

Example (Analysis)**.** $X = C([0, 1], \mathbb{R})$ and $E = \mathbb{P}$. The following fact is true:

$$\forall f \in X \text{ there exists } (P_n) \subseteq E \text{ such that } p_n \to f \text{ uniformly.}$$

Example (Linear Algebra)**.** Let $X = M(n, \mathbb{R})$ and $E =$ Diagonal $n \times n$ real matrices. The connection here is that if $A \in X$ has n linearly independent eigevectors $A = PDP^{-1}$ where P is invertible and $D \in E$. Another connection is that if $A \in X$ is symmetric, then $A = PDP^T$ where P is orthogonal and $D \in E$. (We say that every symmetric matrix is orthogonally diagonalizable)

Example (Complex Matrices)**.** Let $X = M(n, \mathbb{C})$ and $E =$ diagonal $n \times n$ matrices in X. If A is unitary (Hermitian) in X, then $PDP^* = A$ for some unitary matrix P and $D \in E$.

Remark (Linear Algebra)**.** If $A \in M(n, \mathbb{C})$, then A is unitarily diagonalizable if and only if $A^*A =$

AA^* where such matrices are called normal matrices. This tells us that hermitian and unitary matrices are normal in this case.

Definition (Torus Subgroup). Let G be a Lie group. A Lie subgroup H of G is said to be a **Torus subgroup** of dimension $k \geq 1$ if

$$H \cong T^k$$

where $\cong$ is a Lie group isomorphism. That is, H is a $k-$dimensional Torus subgroup if H has dimension k and it is compact, connected, and abelian.

Definition (Maximal Torus). Let G be a Lie group and H be a Torus subgroup of G. We say that H is a **maximal torus** if H is not contained in any larger Torus subgroup of G.

Example ($\mathsf{SO}(2) = \mathsf{U}(1) = S^1$). Clearly, $\mathrm{SO}(2)$ is its own maximal Torus.

Example ($\mathsf{SO}(3)$). Note that

$$H = \left\{ R^z_\theta = \begin{pmatrix} \cos\theta & \sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix} : \theta \in \mathbb{R} \right\}$$

is a copy of $\mathrm{SO}(2) = S^1 = T^1$ inside $\mathrm{SO}(3)$, so it is a Torus subgroup. We claim that H is maximal torus in $\mathrm{SO}(3)$. The reason is as follows: Let T be a Torus subgroup of $\mathrm{SO}(3)$ that contains H. It suffices to show that $T = H$ (and also H is maximal). Since any Torus subgroup is abelian $=$, any $A \in T$ commutes with all $R^z_\theta \in H$:

$$\forall A \in T \text{ and } \forall \theta \in \mathbb{R} \qquad AR^z_\theta = R^z_\theta A.$$

In what follows, we will show that if $A \in \mathrm{SO}(3)$ satisfies (*), then $A \in H$. Let e_1, e_2, e_3 be our standard basis vectors. In order to show that $A \in \mathrm{SO}(3)$ that satisfies (*) is indeed H, it suffices to show that:

(1) $A|_{(e_1,e_2)-\text{plane}}$ is a Linear isometry of the (e_1, e_2) plane.

(2) $\det A > 0$.

Proof. (1) It is enough to show that $Ae_1 \in (e_1, e_2)-$plane and $Ae_2 \in (e_1, e_2)-$plane. Let $Ae_1 = \alpha_1 e_1 + \alpha_2 e_2 + \alpha_3 e_3$. Our goal is to show that $\alpha_3 = 0$. Note that, by assumption, A commutes with all R^z_θ, in particular, it commutes with

$$R^z_\pi = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Therefore,

$$AR^z_\pi(e_1) = R^z_\pi A(e_1) \iff AR^z_\pi(e_1) = A$$

∎

Main point of this lecture is that not all Lie groups have a Torus subgroup.

End of Tuesday Lecture

Remark. We can travel from the Lie Algebra to the Lie group using the exp : $\text{Lie}(G) \to G$ (where $\text{Lie}(G)$ is a Lie Algebra of the Lie group). This tells us that every compact Lie group has a Torus subgroup. Indeed, if x is any nonzero vector in $\text{Lie}(G)$, then the closure of $\{\exp(tx) : t \in \mathbb{R}\}$ is a compact connected abelian Lie subgroup of G. Note that H is a subgroup of $\text{Lie}(G)$ which is based on the two observations:

(1) $e^{tx}e^{sx} = e^{(t+s)x} = e^{sx} \cdot e^{tx}$

(2) $e^{tx}e^{-tx} = e^{0x} = 1_G$.

Note that

(1) If H is a subgroup of Lie group G, then $\overline{H}$ is also a subgroup of G. Furthermore, if this is also abelian (which it is), then the closure is also abelian.

(2) Closure of a connected set is connected (in any topological space).

(3) A closed subset of a compact is compact.

We will prove (1) in this case.

Let G be a Lie group. Let H be an algebraic subgroup of G. Then the closure of H, denoted by $\overline{H}$, is a closed subgroup of G (and so it is a Lie subgroup of G). Moreover, if H is abelian, $\overline{H}$ is abelian also.

Proof. Note that clearly, $\overline{H}$ is closed. It suffices to show that $\overline{H}$ is closed under multiplication. Suppose a and b are in $\overline{H}$. Our goal is to show that $ab \in \overline{H}$. Note that $a \in \overline{H}$, there exists a sequence (a_n) in H such that $a_n \to a$ in G. Since $b \in \overline{H}$, there exists a sequence (b_n) in H such that $b_n \to b$ in G. Therefore,

$$(a_n, b_n) \to (a, b) \text{ in } G \times G.$$

Let's denote multiplication in G by $m : G \times G \to G$. We know $m : G \times G \to G$ is continuous. Then

$$m(a_n, b_n) = m(a, b) \implies a_n b_n \to ab \in G.$$

So, ab is the limit of a sequence of H. Thus, $ab \in \overline{H}$.

Also, $\overline{H}$ contains the inverse of each of its elements. Suppose $a \in \overline{H}$. Our goal is to show that $a^{-1} \in \overline{H}$. We have

$$a \in \overline{H} \implies \exists (a_n) \subseteq H \text{ such that } a_n \to a \in G.$$

Since inversion is a continuous map, we have

$$a_n^{-1} \to a^{-1} \in G.$$

Thus, a^{-1} is the limit of a sequence in H. Thus, $a^{-1} \in \overline{H}$.

Now, we want to prove that $\overline{H}$ is abelian. Suppose $a, b \in \overline{H}$. Our goal is to show that $ab = ba$. Similarly to what we saw above, there exists $(a_n) \subseteq H$ such that $a_n \to a$ in G and there exists a sequence $(b_n) \subseteq H$ such that $b_n \to b$ in G. So, we have $a_n b_n \to ab$ and $b_n a_n \to ba$. Then $ab = ba$. ■

Remark (Fact Used in the Above Proof From Analysis). If $z_n \to z$ and f is continuous, then $f(z_n) \to f(z)$.

6 Week 10

6.1 Tangent Spaces

Let M be a smooth manifold. What do we mean by a **smooth path** in M?

Definition (Smooth Path)**.** A **smooth path** in M is a smooth function $f : I \to M$ with $t \mapsto f(t)$ where $I \subseteq \mathbb{R}$ is a nonempty open interval. (We refer to this path as the path f or the path $f(t)$).

- Each component of the function is a smooth path in $\mathbb{R}$.
- Path connected implies continuous paths between two points a and b.
- We can also extend this notion to smooth paths between a and b.

Example (Examples of Smooth Paths)**.**

- $r : (-1, 1) \to \mathbb{R}^2$ defined by $r(t) = (t, t^2)$ is a smooth path in $M = \mathbb{R}^2$.
- $A : (0, \pi) \to \mathrm{SO}(2)$, where

$$A(t) = \begin{pmatrix} \cos t & -\sin t \\ \sin t & \cos t \end{pmatrix}$$

  is a **smooth path** in $M = \mathrm{SO}(2)$. Consider the following two situtations: Let $f : (0, \pi) \to \mathbb{R}^4$ where $t \mapsto (\cos t, -\sin t, \sin t, \cos t)$ and $A : (0, \pi) \to \mathrm{SO}(2)$ where

$$t \mapsto \begin{pmatrix} \cos t & -\sin t \\ \sin t & \cos t \end{pmatrix}.$$

  Note from these two examples, the implicit assumption is $\mathrm{SO}(2) \subseteq \mathrm{Gl}(2, \mathbb{R}) \subseteq \mathbb{R}^4$.
- (An example of a non-smooth map) Let $A : (-\pi/2, \pi/2) \to \mathrm{SO}(2)$, where

$$A(t) = \begin{pmatrix} \cos t & -\sin t \\ \sin t & \cos t \end{pmatrix}$$

  is not a smooth path in $\mathrm{SO}(2)$. (However, it is a continuous path in $\mathrm{SO}(2)$).

Theorem (Manifold Theory (Relating to the 2nd Example Above))**.** Suppose N, M, L are three smooth manifolds. Suppose M is an embedded sub-manifold of L. Suppose $F : N \to M$. Then $F : N \to M$ is smooth if and only if $i \circ F : N \to L$ where i is the inclusion map ($i : M \to L$)

In our example (2), $N = (0, 1)$ and $M = \mathrm{SO}(2)$ and $L = \mathbb{R}^4$. It is clear that $A : N \to L$ is smooth (because each smooth was smooth). So, by the above claim $A : N \to M$ is also smooth.

Note that derivative of a path can be thought of in the following way:

- If $r(t) = (r_1(t), r_2(t), \ldots, r_n(t))$ is a smooth path in $\mathbb{R}^n$, then

$$r'(t) = \lim_{\delta t \to 0} \frac{r(t + \delta t) - r(t)}{\delta t} = (r_1'(t), r_2'(t), \ldots, r_n'(t)).$$

Remark. Let M be an embdedded submanifold of $\mathbb{R}^k$. Let $p \in M$. As we know the tangent space to M at p, T_pM, can be viewed as a vector space associated with $p \in M$. What is the precise definition of T_pM?

The answer is: There are multiple ways to define T_pM. Here is one of them

- $v \in T_pM$ if and only if there exists a path $A(t)$ in M such that $A(0) = p$ and $A'(0) = p$.

- More precisely, $v \in T_pM$ if and only if there exists $\varepsilon > 0$ and a smooth path $A : (-\varepsilon, \varepsilon) \to M$ such that $A(0) = p$ and $A'(0) = v$.
- Note that $A'(0)$ is sometimes referred to as the velocity vector of the path $A(t)$ at the point p.

Note that T_pM consists of the velocity vectors of all smooth paths in M through $p \in M$.

Example. Prove that

(1) $T_I(\mathrm{O}(2)) = \{A \in M(2,\mathbb{R}) : A^T = -A\} = \{\begin{pmatrix} 0 & -\theta \\ \theta & 0 \end{pmatrix} : \theta \in \mathbb{R}\}$.

(2) $T_I(\mathrm{SO}(2)) = \{A \in M(2,\mathbb{R}) : A^T = -A\}$.

Proof. Here we will prove (2). The proof of (1) is the same.

(I) We claim that $T_I(\mathrm{SO}(2)) \subseteq \mathrm{skew}(2,\mathbb{R})$. Let $X \in T_I(\mathrm{SO}(2))$. This tells us that there exists a smooth path $A : (-\varepsilon, \varepsilon) \to \mathrm{SO}(2)$ such that $A(0) = I$ and $A'(0) = X$. Note that for all $t \in (-\varepsilon, \varepsilon)$ $A(t) \in \mathrm{SO}(2)$. Then for all $t \in (-\varepsilon, \varepsilon)$ $A(t)^T A(t) = I$. Hence, we have

$$\frac{d}{dt}[A(t)^T A(t)] = 0 \Longrightarrow A'(t)^T A(t) + A(t)^T A'(t) = 0.$$

In particular, if $t = 0$, we get

$$\begin{aligned} &A'(0)^T A(0) + A(0)^T A'(0) = 0 \\ &\Longrightarrow X^T I + I^T X = 0 \\ &\Longrightarrow X^T + X = 0 \\ &\Longrightarrow X^T = -X. \end{aligned}$$

Hence, $X \in \mathrm{Skew}(2,\mathbb{R})$.

(II) Now, we show that $\mathrm{Skew}(2,\mathbb{R}) \subseteq T_I\mathrm{SO}(2)$. Let $\begin{pmatrix} 0 & -\theta \\ \theta & 0 \end{pmatrix}$ be an arbitrary elements of $\mathrm{Skew}(2,\mathbb{R})$. We need to show that this matrix is in $T_I\mathrm{SO}(2)$. That is, we need to show that there exists a smooth path $A(t)$ in $\mathrm{SO}(2)$ such that $A(0) = I$ and $A'(0) = \begin{pmatrix} 0 & -\theta \\ \theta & 0 \end{pmatrix}$. Consider the path $A : (-1, 1) \to \mathrm{SO}(2)$ given by

$$A(t) = \begin{pmatrix} \cos(\theta t) & -\sin(\theta t) \\ \sin(\theta t) & \cos(\theta t) \end{pmatrix}.$$

Notice that, clearly, $A(0) = I$. Also,

$$A'(t) = \begin{pmatrix} -\theta\sin(\theta t) & -\theta\cos(\theta t) \\ \theta\cos(\theta t) & -\theta\sin(\theta t) \end{pmatrix}$$

and so we have

$$A'(0) = \begin{pmatrix} 0 & -\theta \\ \theta & 0 \end{pmatrix}.$$

■

Remark. (i) As we saw in part 1 of the proof above, onec an often find a condition that tangent vectors at the identity of a classical matrix Lie group G must satisfy by differentiating the defining equation of G.

(ii) Once this condition is identified, the next question is whether every vector satisfying it actually arises as a tangent vector at the identity. In other words, we need to check whether there exists smooth paths in G whose tangent at the identifies matches the condition (this was part 2 of our solution).

(iii) It is at this stage, when constructing smooth paths in G, that a particularly useful tool, **the matrix exponential**, comes to our aid.

Definition (Matrix Exponential). Let $A \in M(n, \mathbb{C})$. We define the exponential of A, denoted by $\exp(A)$ or e^A, as follows:

$$\exp(A) = \sum_{m=0}^{\infty} \frac{A^m}{m!} = I + A + \frac{A^2}{2!} + \cdots$$

Example.

- Let $D = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$. Compute e^D. Recall that if D is any diagonal matrix, then for every natural number m, D^m is just raising the diagonal entries by m. Then

$$\begin{aligned} e^D &= \sum_{m=0}^{\infty} \frac{D^m}{m!} = \sum_{m=0}^{\infty} \frac{1}{m!} \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}^m \\ &= \sum_{m=0}^{\infty} \frac{1}{m!} \begin{pmatrix} 2^m & 0 \\ 0 & 3^m \end{pmatrix} \\ &= \end{aligned}$$

For the next step, let us be careful when distributing the series in. Note that

$$\sum_{m=0}^{\infty} \frac{1}{m!} \begin{pmatrix} 2^m & 0 \\ 0 & 3^m \end{pmatrix} = \lim_{N \to \infty} \sum_{m=0}^{N} \frac{1}{m!}$$

7 Week 11

7.1 Opening Remarks with Exponential of a Matrix

Remark. For which matrices is it easy to compute the exponential?

(1) Diagonal matrices

(2) Nipoltent matrices (finite sum); that is, matrices where there exists a $k \in \mathbb{N}$ such that $A^k = O_{n \times n}$.

- An example,

$$A = \begin{pmatrix} 1 & 1 \\ -1 & -1 \end{pmatrix}$$

is nilpotent.

- More generally, nilpotent 3×3 matrices have the following form

$$\begin{pmatrix} a & a & a \\ b & b & b \\ (a+b) & -(a+b) & -(a+b) \end{pmatrix}$$

is nilpotent.

(3) Diagonalizable matrices

Proposition (Powers and Exponential of Diagonalizable Matrices). Suppose that $A \in M(n, \mathbb{C})$ is **Diagonalizable**, that is, suppose there exists an invertible matrix P and a diagonal matrix D such that

$$A = PDP^{-1}.$$

Then

(i) For all $m \in \mathbb{N}$ $A^m = PD^mP^{-1}$.

(ii) $\exp(A) = P(\exp(D))P^{-1}$

Proof. (i) For $m = 1$, $A^1 = PD^1P^{-1}$. For $m = 2$,

$$A^2 = AA = PDP^{-1}(PDP^{-1}) = PDDP^{-1} = PD^2P^{-1}.$$

We can do a similar argument by using induction.

(ii) By definition, we have

$$\begin{aligned} e^A = \sum_{m=0}^{\infty} \frac{A^m}{m!} &= \sum_{m=0}^{\infty} \frac{(PDP^{-1})^m}{m!} \\ &= \sum_{m=0}^{\infty} \frac{PD^mP^{-1}}{m!} \\ &= P\Big(\sum_{m=0}^{\infty} \frac{D^m}{m!}\Big)P^{-1} \\ &= Pe^DP^{-1}. \end{aligned}$$

■

Remark (Review of Linear Algebra). If the matrix $A \in M(n, \mathbb{C})$ has n linearly independent eigenvectors, then A is diagonalizable. Indeed,

$$A = PDP^{-1}$$

where

(*) Columns of P are linearly independent eigenvectors of A

(*) Diagonal entries of D are the corresponding eigenvalues.

7.2 Review of Eigenvectors

Example. If $A = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$, the eigenvalues are 4 and 2 which correspond to the (linearly independent) vectors $(1, 1)$ and $(-1, 1)$. Since the number of linearly independent vectors is 2. Then

$$A = PDP^{-1} \text{ where } P = \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \text{ and } D = \begin{pmatrix} 4 & 0 \\ & 0 & 2 \end{pmatrix}$$

where

$$A\begin{pmatrix} 1 \\ 1 \end{pmatrix} = 4\begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

and

$$A\begin{pmatrix}-1\\1\end{pmatrix}=2\begin{pmatrix}-1\\1\end{pmatrix}.$$

Also, it is true tat

$$A=PDP^{-1}$$

where

$$P=\begin{pmatrix}-1&1\\1&1\end{pmatrix},D=\begin{pmatrix}2&0\\0&4\end{pmatrix}.$$

Example. Let

$$A=\begin{pmatrix}2&1&1\\2&3&2\\3&3&4\end{pmatrix}$$

where the eigenvalues and the corresponding eigenvectors are 1 and 7 with $(-1,1,0)$ and $(-1,0,1)$ (corresponding to 1) and 7 corresponding to $(1,2,3)$. Then since there are 3 linearly independent eigenvectors, we have

$$P=\begin{pmatrix}-1&-1&1\\1&0&2\\0&1&3\end{pmatrix}$$

and

$$D=\begin{pmatrix}1&0&0\\0&1&0\\0&0&7\end{pmatrix}.$$

In summary, in order to compute e^X where X is diagonalizable requires the following steps:

(1) Write $X=PDP^{-1}$

(2) Then $e^X=Pe^DP^{-1}$.

7.3 Computing Exponential of Matrices

Example (Skew Symmetric Matrices and its Exponential)**.** Let $\theta\in\mathbb{R}$ be a fixed number. Let $X=\begin{pmatrix}0&-\theta\\\theta&0\end{pmatrix}$. Prove that

$$e^X=\begin{pmatrix}\cos\theta&-\sin\theta\\\sin\theta&\cos\theta\end{pmatrix}.$$

Proof. The corresponding eigenvalues and eigenvectors of this matrix are $-i\theta$ corresponding to $(1,i)$ and $i\theta$ corresponding to $(i,1)$. Check this in private! So, $X=PDP^{-1}$ where

$$P=\begin{pmatrix}1&i\\i&1\end{pmatrix},\ D=\begin{pmatrix}-i\theta&0\\0&i\theta\end{pmatrix}.$$

Note that

$$P^{-1}=\frac{1}{1-i^1}\begin{pmatrix}1&-i\\-i&1\end{pmatrix}=\frac{1}{2}\begin{pmatrix}1&-i\\-i&1\end{pmatrix}=\begin{pmatrix}\frac{1}{2}&\frac{-i}{2}\\\frac{-i}{2}&\frac{1}{2}\end{pmatrix}.$$

Therefore,

$$e^X=Pe^Dp^{-1}=\begin{pmatrix}1&i\\i&1\end{pmatrix}\begin{pmatrix}e^{-i\theta}&0\\0&e^{i\theta}\end{pmatrix}\begin{pmatrix}\frac{1}{2}&\frac{-i}{2}\\\frac{-1}{2}&\frac{1}{2}\end{pmatrix}.$$

Then we have

$$\begin{aligned} e^X &= \begin{pmatrix} e^{-i\theta} & ie^{i\theta} \\ ie^{-i\theta} & e^{i\theta} \end{pmatrix} \begin{pmatrix} \frac{1}{2} & \frac{-i}{2} \\ \frac{-i}{2} & \frac{1}{2} \end{pmatrix} \\ &= \begin{pmatrix} \frac{e^{-i\theta}+e^{i\theta}}{2} & \frac{-ie^{=i\theta}+ie^{i\theta}}{2} \\ \frac{ie^{-i\theta}-ie^{i\theta}}{2} & \frac{e^{-i\theta}+e^{i\theta}}{2} \end{pmatrix} \\ &= \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \end{aligned}$$

where

$$\cos\theta = \frac{e^{i\theta}+e^{-i\theta}}{2} \text{ and } \sin\theta = \frac{e^{i\theta}-e^{-i\theta}}{2i}.$$

Therefore, we have if we have $X = \begin{pmatrix} 0 & -\theta \\ \theta & 0 \end{pmatrix}$, then

$$e^{tx} = \begin{pmatrix} \cos(\theta t) & -\sin(\theta t) \\ \sin(\theta t) & \cos(\theta t) \end{pmatrix} \quad \forall t \in \mathbb{R}.$$

■

Remark. In general, every matrix $X \in M(n, \mathbb{C})$ can be written (uniquely) as

$$X = S + N$$

where S is diagonalizable, N is nilpotent, and $SN = NS$. Then

$$e^X = e^{S+N} = e^S e^N$$

where e^S can be easily computed by diagonalization and e^N can be computed by converting the infinite sum to a finite sum.

Remark. Let G be a Lie group and $e \in G$, be its identity element. Without loss of generality, we may assume that G is an embedded submanifold of $\mathbb{R}^k$ for some $k \in \mathbb{N}$ (so, our construction of T_eG is valid). A question we can ask is "what sort of algebraic operations can we define on T_eG?"

7.4 Operations of Tangent Space at the identity of a Lie Group

Let $X, Y \in T_eG$. Then there exists a smooth path $A(t)$ in G such that $A(0) = e$ and $A'(0) = X$. Likewise, there exists another path $B(t)$ in G such that $B(0) = e$ and $B'(0) = Y$. The first immediate operation we can do with these two paths is to multiply them together which in of itself is a smooth path because the operation is smooth. That is,

$$C(t) = A(t)B(t)$$

is another smooth path in G with $C(0) = e$ and so we have $C'(0) \in T_eG$. In particular, if G is a matrix Lie group, then

$$C'(t) = A'(t)B(t) + A(t)B'(t) \implies C'(0) = A'(0)B(0) + A(0)B'(0) = XI + IY = X + Y.$$

If X and Y are in T_IG, then $X + Y \in T_IG$. Hence, we now know that T_eG is closed under addition.

Now, we will show that this space is closed under scalar multiplication. For any $\alpha \in \mathbb{R}$, $C(t) = A(\alpha t)$ is another smooth path in G with $C(0) = e$. Then $C'(0) \in T_eG$. In particular, if G is a matrix Lie group, then

$$\begin{aligned} C'(t) &= \alpha A'(\alpha t) \\ &= \alpha A'(0) \\ &= \alpha X. \end{aligned}$$

So, we proved that if $X \in T_eG$ and $\alpha \in \mathbb{R}$, then $\alpha X \in T_eG$. Hence, this tells us that T_eG is a linear subspace of $\mathbb{R}^k$.

Note that $C(t) = [A(t)]^{-1}$ is another smooth path in G with $C(0) = e$. That is, $C'(0) \in T_eG$. In particular, if G is matrix Lie group. Then

$$\begin{aligned}
&C'(t) = -A^{-1}(t)A'(t)A^{-1}(t)\\
&\implies C'(0) = -A^{-1}(0)A'(0)A^{-1}(0)\\
&= -IXI\\
&= -X.
\end{aligned}$$

That is, if $X \in T_IG$, then $-X \in T_IG$ (a special case of the last proof).

Next, if we are given $E \in G$ where $C(t) = EB(t)$ is a smooth path in G but $c(0) \neq e$. However, we may consider $C(t) = EB(t)E^{-1}$. Then $C(0) = e$. Indeed,

$$C(0) = EB(0)E^{-1} = EeE^{-1} = EE^{-1} = e.$$

In particular, if G is a matrix Lie group, then

$$\begin{aligned}
&C'(t) = EB'(t)E^{-1}\\
&\implies C'(0) = EYE^{-1}.
\end{aligned}$$

That is, if $Y \in T_Ig$, then for every matrix $E \in G$, $EYE^{-1} \in T_IG$.

For all s, $C_s(t) = A(s)B(t)A(s)^{-1}$ is a smooth path in G and for all s, $C_s(0) = e$. Then for all s, $C_s'(0) \in T_eG$. So, if we let $v(s) = C_s'(0)$, then $v(s)$ is a path inside T_eG. Thus, in particular, $v'(0) \in T_eG$. That is,

$$\frac{d}{ds}\Big|_{s=0}\frac{\partial}{\partial t}\Big|_{t=0} A(s)B(t)A^{-1}(s) \in T_eG.$$

Note that please notice that if G is abelian, then the above expression is 0. Now, suppose G is a matrix Lie group. Let's compute the above vector in T_eG. We have

$$\begin{aligned}
\frac{d}{ds}\Big|_{s=0}\frac{\partial}{\partial t}\Big|_{t=0} A(s)B(t)A^{-1}(s) &= \frac{d}{ds}\Big|_{s=0} A(s)B'(0)A^{-1}(s)\\
&= A'(0)B'(0)A^{-1}(0) + A(0)B'(0)(-A(0)A'(0)A^{-1}(0))\\
&= A'(0)B'(0)A^{-1}(0) - A(0)B'(0)(A(0)A'(0)A^{-1}(0))\\
&= XYI - IY(IXI)\\
&= XY - YX.
\end{aligned}$$

This remarkable equation that we have obtained at the end is called the **Lie Bracket**.

Remark (Observation). Suppose G is an abstract group. Note that G is abelian if and only if for all $x, y \in G$ $xy = yx$. This is equivalent to for all $x, y \in G$, $(xy)(yx)^{-1} = e$ which is further equivalent to for all $x, y \in G$, $xy(x^{-1}y^{-1}) = e$.

Similarly, for all s, $C_s(t) = A(s)B(t)A^{-1}(s)B^{-1}(t)$ is a smooth path in G and for all s, $C_s(0) = e$. Then we have for all s, $C_s' \in T_eG$. Therefore, if we define $v(s) = C_s'(0)$, then $v(s)$ is a path inside the vector space T_eG. Thus, $v'(0) \in T_eG$. That is,

$$\frac{d}{ds}\Big|_{s=0}\frac{d}{dt}\Big|_{t=0} A(s)B(t)A^{-1}(s)B^{-1}(t) \in T_eG$$

Note that if G is abelian, the above expression is zero. In particular, if G is a matrix Lie group, we have

$$\begin{aligned}
\frac{d}{ds}\Big|_{s=0}\frac{\partial}{\partial t}\Big|_{t=0}[A(s)B(t)A^{-1}(s)B^{-1}(t)] &= \frac{d}{ds}\Big|_{s=0}[A(s)B'(t)A^{-1}(s)B^{-1}(t) + A(s)B(t)A^{-1}(s)(-B^{-1}(t)B'(t)B^{-1}(t))]\Big|_{t=0} \\
&= \frac{d}{ds}\Big|_{s=0}[A(s)B'(0)A^{-1}(s)B^{-1}(0) \\
&- A(s)B(0)A^{-1}(s)B^{-1}(0)B'(0)B^{-1}(0)] \\
&= \frac{d}{ds}\Big|_{s=0}[A(s)YA^{-1}(s) - A(s)A^{-1}(s)Y] \\
&= \frac{d}{ds}\Big|_{s=0}[A(s)YA^{-1}(s) - Y] \\
&= \frac{d}{ds}\Big|_{s=0}[A(s)YA^{-1}(s)] \\
&= XY - YX \qquad \text{(See previous proof)}
\end{aligned}$$

Remark. Note that from the previous two cases produce the zero map on T_eG if G is abelian:

$$\forall x, y \in T_eG \quad (x, y) \mapsto 0.$$

We define the **Lie Bracket** on T_eG by for all $X, Y \in T_eG$

$$[X, Y] = \frac{d}{ds}\Big|_{s=0}\frac{\partial}{\partial t}\Big|_{t=0}A(s)B(t)A^{-1}(s)B^{-1}(t).$$

When we equip T_eG with this Lie bracket is called the Lie algebra of g and is denoted by $\mathrm{Lie}(G)$.

7.5 Lie Algebras

Definition (Lie Algebra)**.** A real (or complex) Lie algebra is a real (or complex) vector space L equipped with a function $[\cdot,\cdot] : L \times L \to L$.

(i) $[\cdot,\cdot]$ is bilinear

(ii) $[\cdot,\cdot]$ is skew-symmetric

(iii) $[\cdot,\cdot]$ satisfies the Jacobi identity; that is, for all $x, y, z \in L$, we have

$$[x, [y, z]] + [z, [x, y]] + [y, [z, x]] = 0.$$

One way to create a Lie Algebra is to start with a Lie group and look at the tangent space at the identity of the Lie group and define a Lie Bracket on it.